

Harsh Bordekar

AI Scientist

Engineering-Task Verifiers, Evaluation & Agentic Harnesses for CAE / CAD
PhD Candidate, DLR & TU Braunschweig — Aerospace Composites & Structural Simulation

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🌐 [Harsh's LinkedIn](#)

📍 Braunschweig, Germany

🔗 github.com/harshbordekar-stack

🏠 Permanent Resident - Germany

PROFILE

Simulation and applied-ML engineer with 7+ years of hands-on experience running, debugging, and validating real engineering workflows in CAE (ABAQUS, Nastran/Patran) and CAD (SolidWorks, CREO, AutoCAD), including 3+ years building and evaluating ML/agentic systems on top of that engineering work. PhD work centers on defining what counts as a correct result for a real engineering task: derived a certification-aligned failure criterion for CFRP hydrogen storage tanks, then verified and technically reviewed externally and student-produced simulation models against defined acceptance criteria before sign-off. That is the same task-correctness judgment a verifier needs to encode, just currently applied by a human reviewer rather than to model outputs. Also designed and built a LangGraph-orchestrated agentic AI application that accepts natural-language queries, retrieves context via a hierarchical RAG pipeline, and calls a physics-based CNN as a tool to return homogenized material properties, with retrieval/prediction faithfulness validated against ground-truth simulation results on a held-out set, a concrete evaluation benchmark rather than a one-off demo. Comfortable in Python, Linux, and HPC/SLURM environments day to day, and experienced at explaining simulation and ML concepts to both engineering and non-technical audiences, including instructing 60+ Master's students in FEM/CFRP methods.

Core Competencies: Engineering-Task Verification & Evaluation Criteria Design, Model Verification & Validation (V&V), Agentic AI Systems (LangGraph, Tool-Calling, RAG), CAE Solvers (ABAQUS, Nastran/Patran), CAD (SolidWorks, CREO, AutoCAD), Diagnostic / Explainable AI, Uncertainty Quantification, HPC-Scale Simulation Campaigns (SLURM), Python

PROFESSIONAL EXPERIENCE

Structural Analysis / Simulation Engineer

DLR & TU Braunschweig

Engineering-task verification, evaluation & agentic tooling (PhD)

Feb 2021 - Present

- Verified and technically reviewed externally and student-produced simulation models, meshes, and results against defined methods and acceptance criteria, diagnosing failure modes arising from data gaps or modelling limitations and issuing documented sign-off-style findings. This is direct, repeated practice at defining and applying what "correct" means for an engineering task, beyond surface-level plausibility.
Tools: ABAQUS, Nastran, Python
- Designed and built a LangGraph-orchestrated agentic AI application for microstructure-informed materials-property prediction, accepting natural-language queries and microstructure images, retrieving context via a hierarchical RAG pipeline over **50** materials documents, and calling a physics-based CNN as a tool to return homogenized/effective material properties, replacing full-scale FEM homogenization runs (hours) with on-demand inference (seconds). Retrieval/prediction faithfulness validated at **93%** against ground-truth homogenization results on a held-out set of more than **100** cases, a working evaluation harness for a tool-calling engineering agent, not just the agent itself.
Tools: Python, LangGraph, Physics-Based CNN (Tool-Calling), Hierarchical RAG, LLM Agent Framework
- (PhD) Derived a certification-aligned leakage failure criterion mapping combined strain/load states to leakage onset in CFRP hydrogen storage tanks, bridging progressive-damage models and system-level tank simulations. This defined the physics-based evaluation standard against which a separate multiscale surrogate framework's coverage, accuracy, and failure modes are assessed, the same role a task-specific verifier plays for a model's output.
Tools: ABAQUS, Python, FEM post-processing using C++
- Performed Monte Carlo uncertainty quantification over multiscale simulation outputs to generate leakage-probability-vs-strain relationships, characterising prediction confidence and enabling risk-informed design decisions for next-generation CFRP pressure vessels. This is direct, hands-on experience tracing model outputs back to confidence bounds under uncertainty, the diagnostic half of building an evaluation benchmark.
Tools: Python (SciPy, NumPy), Monte Carlo methods, ABAQUS
- (PhD) Developed a probabilistic, physics-based multiscale surrogate framework for leakage-onset prediction in CFRP hydrogen storage tanks, coupling microstructure-informed mesoscale damage models with stochastic spatial defect distributions to propagate failure across ply, laminate, and structural scales, replacing [x] full-scale FEM runs with a physics-consistent surrogate trained on simulation-generated data.
Tools: Python, ABAQUS, FEM, HPC/SLURM, Multiscale Modelling
- Ran large-scale thermal and structural simulation campaigns on HPC infrastructure to evaluate thermal-protection effectiveness for the Callisto reusable-rocket programme's metal upper housing bay, generating the high-fidelity simulation data such campaigns depend on under production-scale compute constraints.
Tools: ABAQUS, SLURM (HPC), Python, Nastran, Patran

- Instructed 60+ Master's students in FEM theory and ABAQUS-based CFRP modelling, covering Continuum Damage Mechanics (CDM), Progressive Damage Analysis (PDA), and Cohesive Zone Modelling (CZM). This demonstrated the ability to communicate simulation and ML concepts to both technical and non-technical audiences.
Tools: ABAQUS, Automation using Python

Research Intern, Metallic Fatigue & Damage Tolerance

Explainable AI & physics-based diagnostic tooling

DLR-Braunschweig

Nov 2019 - Jan 2021

- Developed a CT-based, explainable-AI defect-detection framework achieving **91%** classification accuracy (**89** precision / **91** recall) on **1200** CT scans of additively manufactured titanium parts, cutting manual review time to **<0.5 sec per image**, with SHAP, Grad-CAM and LIME outputs used as diagnostic tooling to support engineering sign-off. (Published: *Journal of Intelligent Manufacturing*, 2025.)
Tools: Python (scikit-learn, TensorFlow), SHAP, Grad-CAM, LIME, CT image processing

- Assessed fatigue and defect tolerance of aerospace-grade titanium components using small-defect fatigue models (Murakami, El-Haddad), coupling defect measurements with numerical simulation to translate physics-based fracture-mechanics criteria into allowable-defect acceptance limits.

Tools: Small-defect fatigue models (Murakami, El-Haddad), Python (NumPy/SciPy), FEM (ABAQUS)

Student Research Assistant

Institute of Static and Dynamic

Leibniz University Hannover

Sep 2018 - Oct 2019

- Developed an ML framework for automated detection and quantification of fiber undulations in unidirectional CFRP laminates from image data, generating spatially resolved defect metrics used directly as training/validation inputs for probabilistic structural simulation models assessing manufacturing-induced variability.

Tools: MATLAB, Python (scikit-learn, TensorFlow), image processing, NumPy/SciPy

Design Engineer

CAD design & manufacturing quality assurance

Gujarat Industrial Development Corporation, AK. Alloys

Mar 2017 - Mar 2018

- Designed sand-cast components in CAD, translating functional requirements into production-ready geometries compliant with sand-casting process limitations, draft angles, parting-line constraints, and dimensional quality standards. Managed witness casting and quality-assurance checks against engineering specifications and customer acceptance criteria, real, hands-on CAD-to-manufacturing workflow experience predating the simulation/ML work above.

Tools: CREO, SolidWorks, AutoCAD

SKILLS SUMMARY

Engineering-Task Verification & Evaluation	Verification & Validation (V&V) Acceptance-Criteria Definition Physics-Based Evaluation Benchmarks Failure-Mode Diagnosis Uncertainty Quantification (Monte Carlo)
Agentic Systems & LLM Tooling	LangGraph Retrieval-Augmented Generation (RAG) Model Context Protocol (MCP) Tool-Calling LangChain
CAE Solvers & HPC	ABAQUS Nastran/Patran Linear & Non-linear FEM HPC / SLURM (Large-Scale Simulation Campaigns) Multiscale / Progressive Damage Modelling
CAD	SolidWorks CREO AutoCAD Design-for-Manufacture (Sand Casting)
ML & Diagnostic AI	PyTorch TensorFlow Scikit-learn Explainable AI (SHAP, Grad-CAM, LIME) Physics-Based CNN Surrogates
Programming Languages	Python C++ FORTRAN MATLAB
Data Analysis & Scientific Computing	NumPy Pandas Matplotlib Seaborn Plotly SALib SciPy
APIs & Backend Development	RESTful APIs FastAPI JSON
Version Control & Reproducibility	Git Docker Linux
Application Development	VS Code JupyterLab Google Colab Streamlit Gradio
Languages	English (C1) German (B1)

AWARDS AND EDUCATION

 2nd Prize: Best Paper Award	DLR (German Aerospace Center) 2026
 3rd Prize: Best Paper Award	DLR (German Aerospace Center) 2024
PhD Candidate, Computational Methods in Engineering	DLR & TU Braunschweig, Germany Feb 2021 – 2026 (expected)
Dissertation: Physics-Informed, Multiscale Machine Learning for CFRP Composite Structures	
M.Sc. Computational Methods in Engineering	Leibniz University Hannover, Germany April 2018 – Dec 2020
Focus: Numerical Methods, Scientific Machine Learning, Deep Learning	

SELECTED PUBLICATIONS

1. **Bordekar, H.**, et al. “Microstructure-informed Probabilistic Multiscale Modelling of CFRP via Machine-Learning-Derived Stochastic Volume Elements.” *Manuscript in preparation*, 2026.
2. **Bordekar, H.**, et al. “MicroStructAgent: A Physics-Informed Agentic AI Framework for Computational Homogenization and Quality Assurance of Fiber Composite Laminates.” *Manuscript in preparation*, 2026.
3. **Bordekar, H.**, et al. “Microstructure Characterization of Fiber Composite Laminate Using eXplainable Artificial Intelligence.” *Journal of Composite Materials*, vol. 60, no. 1, 2026, pp. 3–25. [DOI](#)
4. **Bordekar, H.**, et al. “eXplainable Artificial Intelligence for Automatic Defect Detection in Additively Manufactured Parts Using CT Scan Analysis.” *Journal of Intelligent Manufacturing*, vol. 36, no. 2, 2025, pp. 957–974. [DOI](#)

REFERENCES

Prof. Dr. Christian Huehne	Abteilungsleiter Funktionsleichtbau. Deutsches Zentrum für Luft- und Raumfahrt. Institut für Systemleichtbau
Prof. Dr. Oliver Voelkerink	Head of Institute., IMA, TU-Braunschweig
Dr. Nicola Cersullo	R&T Project Manager @ Airbus Hamburg